Jatin Kumar Sinha

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SKILLS

Languages: Java, Python, SQL, JavaScript

Backend & APIs: Axios, Spring Boot, FastAPI, REST APIs, WebClient, JWT Auth, WebSockets

Frontend: React, Vite, HTML, CSS, Tailwind CSS, Framer Motion

Databases, Caching & Protocols: PostgreSQL, Redis, Caffeine Cache, TCP/IP, Java Sockets, RESP2

DevOps & Tools: Docker, Git/GitHub, Vercel, Render, Railway, VS Code

Data & ML: NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, Power BI, Excel

Soft Skills: Problem-Solving, Team Player, Project Management, Adaptability

PROJECTS

Mini Redis in Java | Link October 2026

- Built a Redis-like TCP server in Java that parses RESP2 and supports string, list, counter, and key-expiration commands.
- Added persistent snapshots with restart recovery, plus primary–replica synchronization, write propagation, and acknowledgement offsets.
- Created assertion-based integration checks for command handling, persistence, read-only replica behavior, and replication recovery.

Tech used: Java 21, TCP sockets, RESP2, Java I/O, snapshots, virtual threads, Git/GitHub.

F1 Intelligence Hub | Link June 2026

- Built a full-stack F1 analytics dashboard with a React/Vite frontend and FastAPI backend, using the Jolpica API for season-specific driver standings, constructor standings, and race schedules.
- Added season selection across the dashboard, driver profiles, and driver comparisons, along with a historical champion archive.
- Implemented asynchronous API requests, expiring response caching, and loading and error states; deployed the frontend on Vercel and backend on Render.

Tech Used: Python, FastAPI, React, JavaScript, Vite, Axios, Jolpica API, Render, Vercel

Real-Time AQI Alert System using Java | Link Mar 2026

- Built a Java-based air quality monitoring system that fetches real-time AQI data from the OpenWeather API and flags unsafe pollution conditions.
- Implemented JSON parsing, AQI analysis logic, and automated email alerts using the JavaMail API and SMTP integration.
- Enabled real-time pollution monitoring and instant notification delivery, improving awareness and response to hazardous air quality levels.

Tech Used: Java, OpenWeather API, JSON Parsing, JavaMail API, SMTP

TRAINING

Lovely Professional University | Link Jun 2025 – Jul 2025

Tech Used: Python, NumPy, Pandas, Scikit-Learn, Matplotlib, Seaborn, NLP

- Completed an intensive Summer Training Program in AI & ML, applying machine learning techniques to real-world problems.
- Gained hands-on experience in supervised and unsupervised learning using Linear Regression, KNN, Decision Trees, and Clustering.
- Performed data preprocessing (scaling, encoding, missing-value handling) and built models with NumPy, Pandas, and Scikit-Learn, visualized using Matplotlib and Seaborn.
- Developed an interactive mental-health chatbot using NLP techniques (text preprocessing, tokenization, sentiment analysis) to interpret user input and generate supportive responses.

CERTIFICATIONS

Oracle Cloud Infrastructure Certified AI Foundations Associate | Link AUG 2026

Oracle AI Database Certified Foundations Associate | Link AUG 2026

ChatGPT-4 Prompt Engineering: ChatGPT, Generative AI & LLM | Link AUG 2025

Computational Theory: Language Principle & Finite Automata Theory | Link AUG 2025

EDUCATION

Lovely Professional University Phagwara, Punjab
Bachelor of Technology Aug 2023 – Present
Computer Science and Engineering; CGPA:6.73

Bishop Scott Boys' School Patna, Bihar
Intermediate Mar 2021 – May 2022
PCM; Percentage: 72%

Bishop Scott Boys' School Patna, Bihar
Matriculation Mar 2019 – May 2020
Percentage: 75%